



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division

MTech in Artificial Intelligence and Machine Learning

S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS4 - Defence Drone - [Weightage 12%]

Story / Background

A defense organization deploys the autonomous agent which is an unmanned aerial vehicle (UAVs) to locate and intercept the target. Target is the external signal source entered into the restricted environment. The agent must navigate through the environment, containing strict No-Fly Zones (N) and unpredictable Hazardous Weather Regions (W). The objective of the agent is to compute an efficient final path from the start node to the target node using Greedy Best-First Search (GBFS), minimizing the heuristic value.

Problem Statement

Design an intelligent agent using Greedy Best-First Search (GBFS) to navigate from the Start node (S) to the Goal node while avoiding obstacles.

Agent moves and Goals:

- The agent is deployed on-site for continuous monitoring to locate the external signal source and remove it from the environment.
- The agent starts at the node (0,0) and should reach the target at (6,7).
- The agent is intelligent enough to implement the heuristics such that it can reach the target node (E) with the minimum heuristic value.
- Agent can move in orthogonal directions only
 - North(Up), South (Down) East(Right), West(Left).
- If multiple nodes have equal heuristic values, then the priority order to choose the next node is
 - North
 - East

- South
- West

Environment Representation

To ensure evaluation uniformity across all student submissions, the airspace is modeled as a discrete 8×8 2D grid matrix.

	0	1	2	3	4	5	6	7
0	S	.	.	.	N	.	.	.
1	.	W	.	.	N	.	.	.
2	.	.	.	W	N	.	W	.
3	N	N	.	.	.	.	N	.
4	.	.	.	W	.	.	.	.
5	.	.	.	.	.	N	N	.
6	.	.	W	.	.	.	.	E
7	.	.	.	W	.	W	.	.

Legend:

- **Start Node (S):** (0,0) — Agent start node [Score: +2]
- **Goal Node (E):** (6,7) — External Signal Source [Score: +2]
- **Passable Airspace (.):** Standard, low-risk open airspace. [Score: +1]
- **Weather Hazards (W):** Penalty for the agent. . [Score: +4]
- **No-Fly Zones (N):** The agent cannot pass through the no-fly zones.. [Score: +8]
- **Movement Model:** The drone can move in **4 orthogonal directions** (North, South, East, West). Diagonal flight paths are strictly prohibited.

Suggested Heuristics

Solve the problem using the below heuristics

Heuristic (h1): Euclidean distance

$$h_1(n) = \sqrt{(x_n - x_i)^2 + (y_n - y_i)^2}$$

Where, x_n and y_n are the current nodes

x_i , y_i are the current nodes.

Heuristic (h2): Bounding-Box Risk-Weighted Heuristic (future-aware)

The agent is getting aware of the future path by choosing the bounded box weighted heuristic. It is the product of Manhattan distance and the average distance from the current node to the goal node. The agent adds up the node value (1,4,8) from the current node to the target node from all the possible immediate next nodes. The agent in turn calculates the number of bounded possible nodes from the current node to the target node(k) and averages with the sum of the node values from the current node the the destination node.

$$h_2(n) = (|x_n - x_i| + |y_n - y_i|) \times \left(\frac{1}{K} \sum_{\text{cell} \in \text{Box}(n,I)} c(\text{cell}) \right)$$

Example: Agent is at node (3,2) and evaluating the next node and the with the future-aware heuristic (h2)

The agent can choose the next node as (3,3) or (4,2). Calculation for (3,3) is shown below.

- Step 1: Calculate Manhattan Distance to Goal: $|3-6| + |3-7| = 3+4=7$
- Step 2: Define the Bounding Box Window
 - Calculating the sum of nodes from the current node or cell (c),
 - The rectangle spans from Row 3 to 6 and Column 3 to 7. This creates a 4×5 window containing K=20 nodes(cells).
 - $K = ((6-3)+1)((7-3)+1) = 4*5=20$
 - Let's sum up the exact transition costs (c) inside this specific 4×5 sub-grid
 - Row 3: (3,3)=1, (3,4)=1, (3,5)=1, (3,6)=8 [N], (3,7)=1 → Sum = 12
 - Row 4: (4,3)=4 [W], (4,4)=1, (4,5)=1, (4,6)=1, (4,7)=1 → Sum = 8
 - Row 5: (5,3)=1, (5,4)=1, (5,5)=8 [N], (5,6)=8 [N], (5,7)=1 → Sum = 19
 - Row 6: (6,3)=1, (6,4)=1, (6,5)=1, (6,6)=1, (6,7)=1 [I] → Sum = 5
 - Total Window Cost=12+8+19+5=44
- Step 3: Compute Final h2 for Node A
- Average Window Cost=44/20=2.20
- $h_2(3,3)=\text{Distance} \times \text{Average Window Cost} = 7 \times 2.20 = 15.40$

Expectations:

1. Implement the given problem using GBFS with the appropriate data structure
2. The program should
 - a. Have the executable code without any issues
 - b. Implement both the heuristics

- c. Construct initial grid and visual representation of the same. You can choose the appropriate icons for agent, no-fly zones and, weather hazard zones.
 - d. Frontier nodes at each iteration
 - e. Explored nodes
 - f. Heuristic values
 - g. Selected next node
 - h. Final path to reach the target
 - i. Number of no-fly zones and weather hazard zones crossed. This should show the penalties incurred.
3. Complexity Analysis: Visual representation is preferred where possible.
- a. provide the nodes explored,
 - b. time complexity in the Big O notation
 - c. execution time
 - d. memory usage
 - e. provide the analysis for both the heuristics.
4. Fill the below table.

Metric	Description
Nodes Expanded	Total explored nodes
Runtime	Execution time (ms)
Memory Usage	OPEN + CLOSED size
Total Path Cost	Sum of transition costs
Path Length	Total moves
Heuristic	Heuristic value

5. Heuristic function Analysis
- a. Describe which heuristic is better in terms of reaching the target.
 - b. Visual representation of the heuristic values to reach the target.
 - c. Visual representation of the heuristic values vs time to reach the target.

6. Identify situations where the agent becomes trapped or explores inefficient regions. Show the instances in a tabular format containing heuristic (h_1 or h_2), heuristic value, trapped at node, next iteration to come out of the trap.
 - a. NOTE: A trap could be a dead-end or infinite loop.
7. Implement and compare the problem with A* using heuristic 1.
 - a. Compare which algorithm is complete
 - b. Compare the nodes expanded using GBFS and A* with heuristic 1.
 - c. Compare which algorithm is fast enough to reach the target.
 - d. Provide the comparison in a table format or a chart.
8. Explain the PEAS components.
9. Documentation
 - a. Document the program clearly.
 - b. Provide the execution flow or sequence diagram
10. Output of the program should be clear
 - a. Visual representation is required for each survivor.
 - b. Update the initial grid and visually represent the updated grid for the next survivor.
11. Complexity Analysis: Visual representation is preferred where possible.
 - a. provide the nodes explored,
 - b. time complexity in the Big O notation
 - c. execution time
 - d. memory usage
12. Conclusion
 - a. Provide your analysis on the performance of the GBFS algorithm in terms of optimality and completeness.

Deliverables:

- PDF document **designPSXX_<group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Create a single notebook. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:

- **[Group id] _A1_PSXX_XXXXXXXXXX.zip** and submit the zipped file.
- Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries in [this dedicated discussion section](#). Beware that only hints will be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **8th June 2026 11:55 PM IST (Monday).**

2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 8th June 2026 11:56PM to 9th June 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 9th June 2026 11:56PM to 10th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - c. Beyond 10th June 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id] _A1_PSXX_XXXXXXXXXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **12 Marks**.
- Grading will depend on:
 - Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the

value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.